(Journal Resume)

Analisis transkriptom dari respon kekurangan fosfat memberikan wawasan tentang biosintesis strigolakton pada padi

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Pendahuluan

Tumbuhan mengembangkan adaptasi morfologis, fisiologis, biokimia, dan molekuler terhadap defisiensi unsur fosfor (P). Respon utama tumbuhan terhadap hal ini berupa akar yang cenderung lebih banyak tumbuh di permukaan tanah dan peningkatan eksudasi pelarut P dari akar, dan induksi transporter P seperti, P transporter (PHT), protein SYG1/PHO81/XPR1 (SPX), *plastidic phosphorylated compounds/p antiporters*, dan transporter P mirip SULTR. Strigolakton (SL) menjadi hormon yang paling terakumulasi saat defisiensi P yang bertindak mengatur perkembangan tumbuhan dan interaksi rizosfer. Penelitian menggunakan tanaman padi menunjukkan bahwa biosintesis SL melibatkan jalur karotenoid melalui enzim D27, CCD7, dan CCD8. Melalui analisis transkriptomik (RNA-seq) pada mutan d10, ditemukan bahwa SL mengoordinasikan respons tanaman terhadap stres P. Studi ini juga mengidentifikasi dua gen *methyl transferase* baru yang mengkatalisis konversi asam carlactonoat (CLA) menjadi metil carlactonoat (MeCLA). Penelitian ini membandingkan defisiensi P dengan SL sintesis (GR24) melalui metode transkriptomik temporal dengan RNA *sequencing*.

Metode

Bahan Tanaman

Digunakan benih padi varietas Shiokari tipe liar dan mutan biosintesis SL d10. Benih disterilisasi permukaan dan ditumbuhkan dalam larutan Hoagland cukup (+P) selama 7 hari di ruang terkontrol. Bibit dipindahkan ke larutan defisiensi P (-P) dan dipanen kembali setelah 1, 3, 7, dan 8 setelah dipindahkan. Penambahan kembali P dilakukan pada hari ke-7, bibit +P dan -P diberi perlakuan 5um RAC-GR24 selama 1 hari.

Isolasi RNA Total dan Persiapan Pustaka

Akar diekstraksi dengan kit TruSeq Stranded mRNA HT (RS-122-2103) dan sekuensing menggunakan Illumina HiSeq2500 untuk memetakan transkriptom akar dan Bioanalzer Agilent 2100 untuk pengecekkan pustaka.

Pemetaan Pembacaan RNA-seq dan Analisis Ko-ekspresi

Data dipetakan ke genom referensi padi (*Nipponbare*) menggunakan alat seperti STAR dan DESeq2 untuk mencari gen yang diekspresikan secara berbeda (DEG). Peneliti menggunakan gen biosintesis SL yang sudah diketahui (OsCCD7, OsCCD8, MAX1-Os900) sebagai panduan untuk menemukan gen baru yang memiliki pola ekspresi serupa. Selanjutnya dilakukan analisis pengayaan GO dengan g: Profiler dan analisis RT-PCR secara kuantitatif menggunakan iScript cDNA.

Ekspresi Sementara pada *Nicotiana benthamiana*

Gen Arabidopsis (AtD27, AtMAX3, AtMAX4, AtMAX1 dikloning. Panjang penuh dari dua gen kandidat padi (*Os01g0700300* dan *Os01g0701700*) diamplifikasi. Gen padi tersebut dimasukkan ke dalam daun tembakau (*N. benthamiana*) menggunakan bantuan bakteri *Agrobacterium tumefaciens*.

Analis strigolakton

Daun yang telah diinfiltrasi kemudian diekstraksi untuk dianalisis kandungan SL-nya (seperti CL, CLA, dan MeCLA) menggunakan teknologi LC–MS/MS (*Liquid Chromatography–tandem Mass Spectrometry*).

Hasil dan Pembahasan

1. Dinamika Respons Transkriptomik terhadap P

Data RNA-seq menunjukkan bahwa tanaman padi sangat sensitif terhadap status fosfor:

- **Efek Bertahap:** Jumlah gen yang diekspresikan secara berbeda (DEG) meningkat tajam seiring lamanya kelaparan P (dari 77 gen pada hari ke-1 menjadi 1.451 gen pada hari ke-8).
- **Pemulihan Cepat:** Hanya dalam **1 hari pemberian kembali P (RP)**, profil ekspresi gen yang tadinya merespons kekurangan P langsung berbalik kembali ke kondisi normal (kontrol). Ini membuktikan kontrol genetik yang sangat ketat terhadap homeostasis P.
- **Gen Kunci:** Gen seperti *PT* (transporter fosfat), *SPX* (sensor P), dan *PAPs* (fosfatase asam) menunjukkan peningkatan ekspresi yang sangat kuat saat kekurangan P.

2. Interaksi Strigolakton (SL) dan Kekurangan P

Penelitian ini berhasil memetakan bagaimana SL terlibat dalam respons kelaparan P:

- **Umpan Balik Negatif:** Saat P kurang, ekspresi gen biosintesis SL (*D27, CCD7, CCD8, MAX1*) meningkat. Namun, ketika tanaman diberi **GR24** (SL sintetis), ekspresi gen-gen tersebut ditekan kembali. Hal ini menunjukkan adanya mekanisme kontrol internal agar produksi SL tidak berlebihan.

- **Interaksi Hormonal Lain:** Kekurangan P juga memengaruhi jalur karotenoid (prekursor SL dan ABA). Menariknya, SL ditemukan menekan gen *GA2ox2* (katabolisme Giberelin), yang menyiratkan bahwa SL membantu menjaga kadar Giberelin (GA) aktif untuk mengatur pertumbuhan anakan padi.

3. Penemuan Gen Baru dalam Biosintesis SL

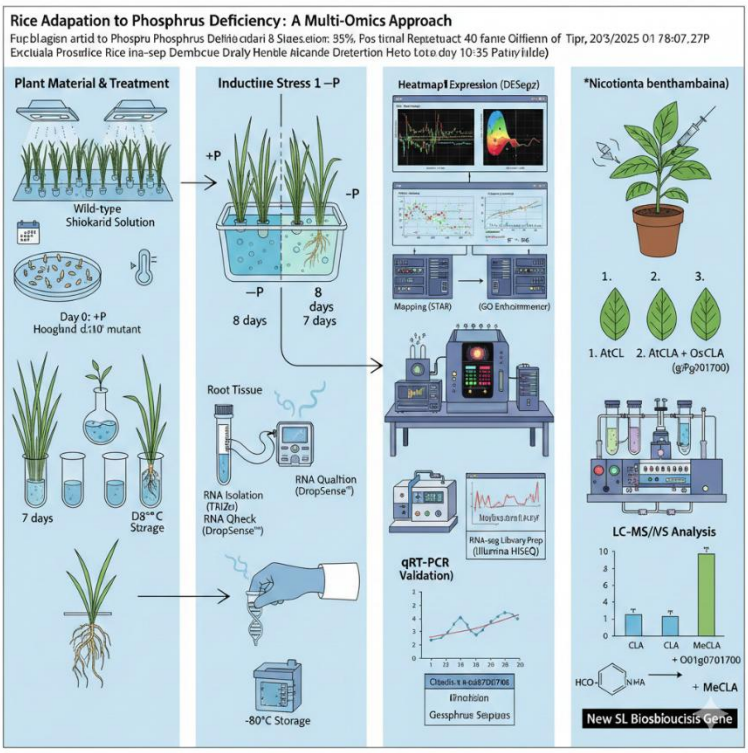
Salah satu temuan paling signifikan dalam bagian ini adalah identifikasi jalur biosintesis **SL non-kanonik**:

- **Analisis Ko-ekspresi:** Dengan mencari gen yang pola kerjanya mirip dengan gen SL yang sudah diketahui, peneliti menemukan dua gen kandidat: **Os01g0700300** dan **Os01g0701700**.
- **Fungsi Metil Transferase:** Melalui uji pada tanaman tembakau (*N. benthamiana*), terbukti bahwa kedua gen ini berfungsi sebagai **CLA METHYL TRANSFERASE**.
- **Hasil Reaksi:** Keduanya mengubah asam carlactonoat (CLA) menjadi **metil carlactonoat (MeCLA)**. Dari keduanya, *Os01g0701700* menunjukkan aktivitas yang lebih tinggi.

Kesimpulan

Gen-gen yang mengatur karotenoid, hormon ABA, dan Strigolakton (SL) saling bekerja sama untuk menjaga keseimbangan nutrisi di dalam tubuh padi. Lewat data yang dikumpulkan, peneliti berhasil menemukan gen kandidat baru untuk pembuatan Strigolakton.

Ilustrasi



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